

Olist E-Commerce Dataset

Product Analysis Report

Top Selling | Revenue Performance | Customer Demand | Geographic Analysis | RFM Segment Preferences

Prepared by Data Analyst | Industry-style Analytical Report

Executive Summary

Total Dataset Rows 82,365 orders | 39 columns
Total Product Categories 73 unique categories analysed
Top Selling Category Bed Bath Table — 5,473 orders
Top Revenue Category Watches Gifts — R\$ 743,296.92
Lowest Selling Category Security And Services — 2 orders
Champions' Top Category Health Beauty — 65 purchases
Loyal Customers' Top Category Health Beauty — 89 purchases
RFM Integration Merged with RFM_Customer_Segmentation.xlsx for segment-level product analysis

Watches Gifts generates R\$743K revenue but ranks only 8th in orders — a premium, high-revenue-per-order category requiring a different strategy than high-volume low-margin products.

1. Scope, Objective and Analytical Approach

Objective: Perform a comprehensive product performance analysis on the Olist e-commerce dataset to identify top-selling and top-revenue categories, understand customer demand by geography, and map product preferences across RFM behavioural segments. The analysis covers five analytical phases: product performance, monetary analysis, customer analysis, geographic demand, and segment-level product preferences.

Data Sources: (1) analysis_data.csv — the cleaned Olist master dataset with 82,365 rows and 39 columns, including order, product, customer, payment, and delivery information; (2) RFM_Customer_Segmentation.xlsx — the 6,263-customer RFM file containing CustomerID, RFM scores, and Segment labels, merged on customer_unique_id for segment-level product analysis.

Revenue column used: price + freight_value per order line item. The product_category_name_english column was used for all category-level groupings. Categories labelled 'Unknown' (uncategorised products) are noted separately and excluded from segment preference analyses where appropriate.

2. Dataset Overview

Table 1. Dataset Summary

Metric	Value	Notes
Total Rows	82,365	All orders including all statuses
Total Columns	39	Including engineered features
Product Categories	73 categories	Includes 'Unknown' for unclassified products
Revenue Column	price + freight_value	Total customer payment per line item
Unique Customers	~69,000+	customer_unique_id level
States Covered	27 Brazilian states	Used in geographic demand analysis
RFM File Merged	6,263 customers	RFM_Customer_Segmentation.xlsx

3. Phase 1 — Product Performance Analysis

3.1 Top 10 Selling Product Categories

The top-selling categories are ranked by the number of unique orders placed in each category. This identifies the highest-demand products on the Olist platform by transaction volume.

Table 2. Top 10 Selling Product Categories by Order Count

Rank	Product Category	Orders
1	Bed Bath Table	5,473
2	Health Beauty	5,089
3	Sports Leisure	4,454
4	Computers Accessories	3,880
5	Furniture Decor	3,731
6	Housewares	3,447
7	Watches Gifts	3,287
8	Telephony	2,430
9	Auto	2,260
10	Toys	2,010

Note: 'Unknown' category (14,600 orders) is excluded from this ranked table as it represents unclassified products.

Figure 1. Top 10 Selling Product Categories



Figure 1: Bed Bath Table leads with 5,473 orders, closely followed by Health Beauty (5,089). These two categories together account for approximately 13.5% of all classified orders.

Bed Bath Table and Health Beauty are the clear market leaders by volume. The top 5 categories — Bed Bath Table, Health Beauty, Sports Leisure, Computers Accessories, and Furniture Decor — collectively drive over 22,000 orders, making them the commercial backbone of the Olist product catalogue.

3.2 Bottom 10 Selling Product Categories

Table 3. Bottom 10 Selling Product Categories by Order Count

Rank	Product Category	Orders
1 (Lowest)	Security And Services	2
2	Fashion Childrens Clothes	6
3	La Cuisine	7
4	Cds Dvds Musicals	8
5	Diapers And Hygiene	13
6	Fashion Sport	15
7	Home Comfort 2	16
8	Arts And Craftmanship	16
9	Flowers	17
10	Fashio Female Clothing	20

Figure 2. Bottom 10 Selling Product Categories



Figure 2: Security And Services (2 orders) and Fashion Childrens Clothes (6 orders) are virtually non-existent on the platform. These categories may represent catalogue gaps or very niche offerings.

The bottom 10 categories show extremely low order volumes — all under 20 orders. These may represent niche products with limited demand, categories that are not well-merchandised on the platform, or product types that customers prefer to purchase through other channels. Strategic catalogue review for these underperforming categories is warranted.

4. Phase 2 — Revenue Analysis

4.1 Top 10 Revenue Generating Categories

Revenue is calculated as price + freight_value per order line item, summed across all orders in each category. This identifies which categories drive the most commercial value — which may differ significantly from order volume rankings.

Table 4. Top 10 Revenue Generating Product Categories

Rank	Product Category	Total Revenue (R\$)
1	Watches Gifts	743,296.92
2	Health Beauty	737,409.81
3	Bed Bath Table	625,785.69
4	Sports Leisure	590,918.46
5	Computers Accessories	540,672.77
6	Furniture Decor	440,672.31
7	Housewares	392,340.16
8	Cool Stuff	381,264.17
9	Auto	360,329.79
10	Garden Tools	292,393.44

Figure 3. Top 10 Revenue Generating Product Categories

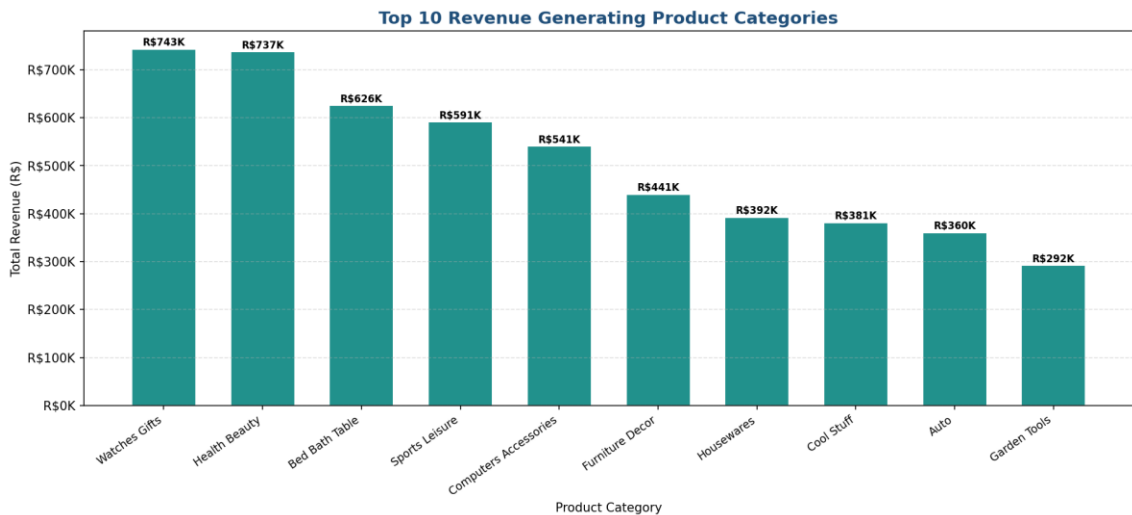


Figure 3: Watches Gifts (R\$743K) and Health Beauty (R\$737K) are nearly tied for the top revenue position. Together they account for approximately 18.8% of total classified product revenue.

A key strategic insight emerges when comparing revenue rankings to order volume rankings: Watches Gifts ranks 7th by orders but 1st by revenue, indicating a significantly higher average order value. This is a premium product category where individual transactions carry higher commercial value. Conversely,

Bed Bath Table ranks 1st by orders but 3rd by revenue, indicating a high-volume, moderate-value category.

4.2 Bottom 10 Revenue Generating Categories

Table 5. Bottom 10 Revenue Generating Product Categories

Rank	Product Category	Total Revenue (R\$)
1 (Lowest)	Security And Services	283.29
2	Fashion Childrens Clothes	419.86
3	Cds Dvds Musicals	465.00
4	Home Comfort 2	618.37
5	Flowers	673.55
6	Diapers And Hygiene	864.79
7	Fashion Sport	1,034.52
8	Arts And Craftmanship	1,195.34
9	La Cuisine	1,378.98
10	Fashio Female Clothing	1,568.17

Figure 4. Bottom 10 Revenue Generating Product Categories

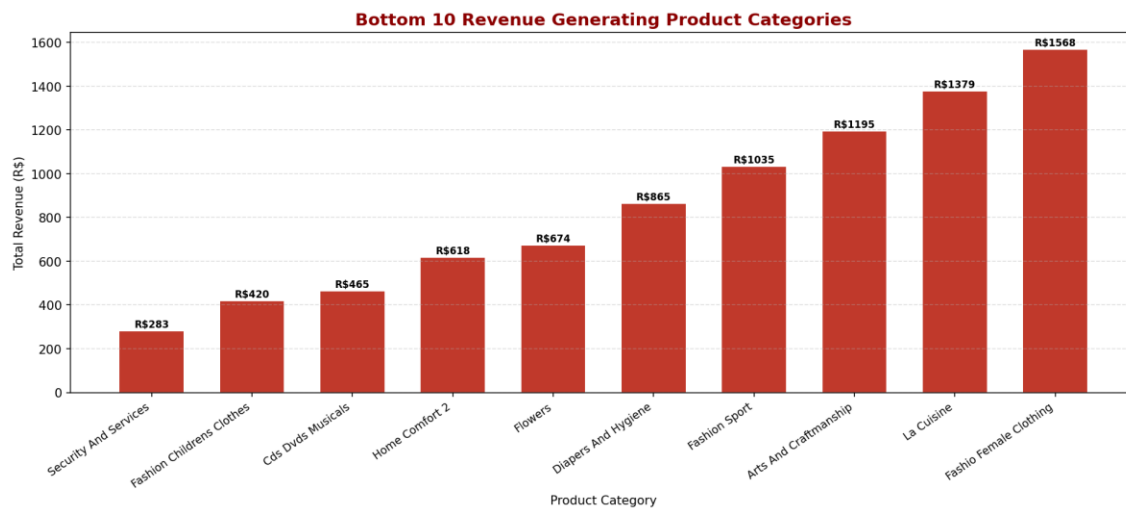


Figure 4: Security And Services (R\$283) and Fashion Childrens Clothes (R\$420) generate negligible revenue. All 10 categories generate under R\$1,600 in total revenue.

4.3 Revenue Contribution — Top 10 Categories

Figure 5. Revenue Contribution Pie — Top 10 Product Categories

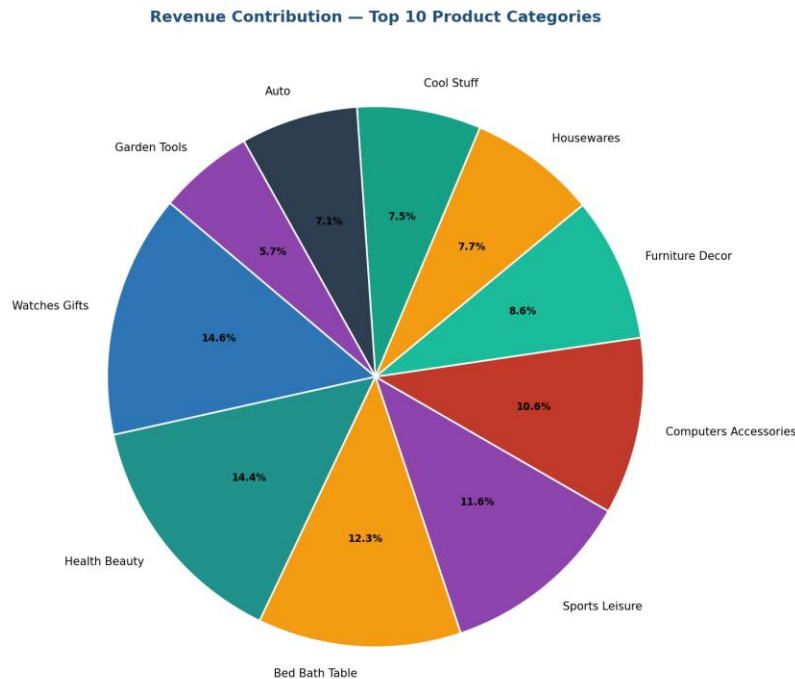


Figure 5: Watches Gifts and Health Beauty together account for approximately 37.7% of top-10-category revenue. The top 10 categories represent a concentrated revenue base.

4.4 Orders vs Revenue Comparison

Figure 6. Top 10 Categories — Orders vs Revenue Comparison

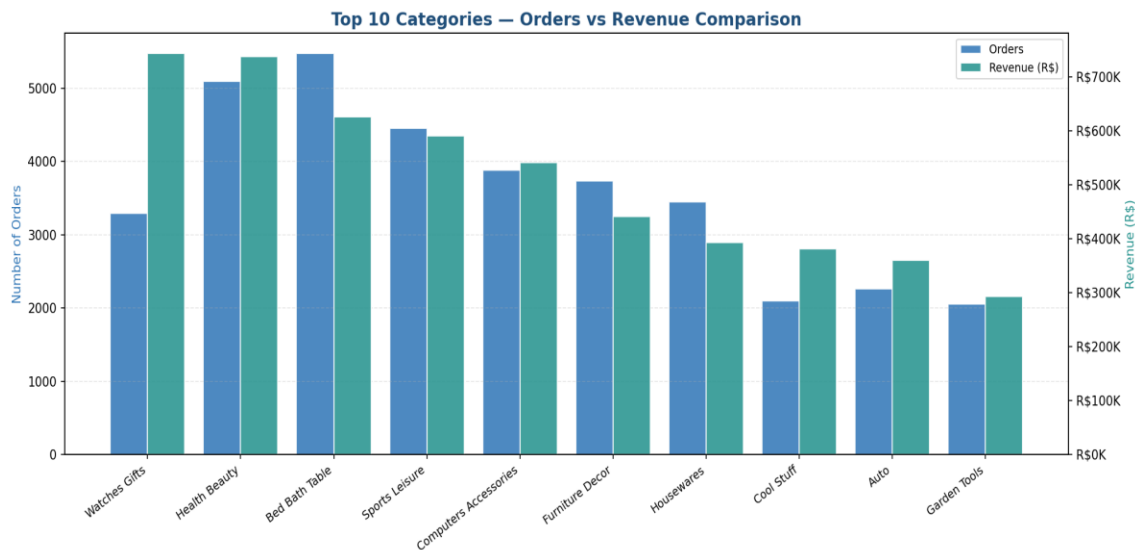


Figure 6: Dual-axis chart comparing order counts (blue bars, left axis) and revenue (green bars, right axis) for the top 10 revenue categories. Watches Gifts has high revenue relative to its order count, confirming high average order value.

The divergence between order volume and revenue across categories reveals three strategic tiers: (1) High Volume + High Revenue categories (Health Beauty, Bed Bath Table) that are both popular and commercially valuable; (2) Premium categories (Watches Gifts, Cool Stuff) with fewer orders but high revenue per transaction; and (3) High-volume moderate-revenue categories (Computers Accessories, Housewares) that drive traffic but require margin management.

4.5 Average Order Value by Product Category

Figure 7. Top 10 Product Categories by Average Order Value

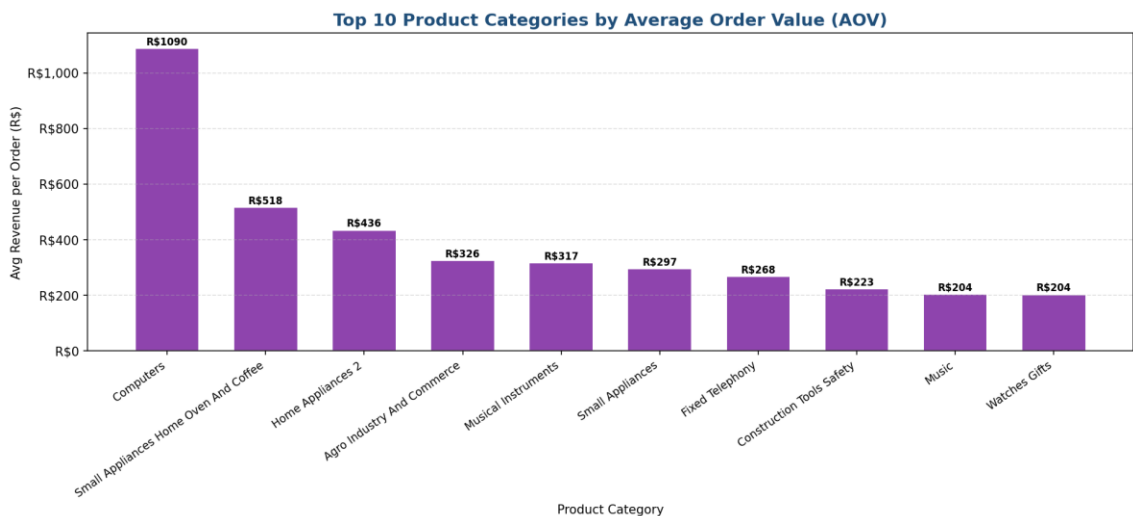


Figure 7: Average revenue per order by category (excluding Unknown). Premium categories appear here even if they have low total order volumes.

Average Order Value (AOV) reveals product categories where individual purchases carry the highest value. Categories with high AOV but moderate volume (such as computers, musical instruments, and watches) are premium segments where fewer customers spend significantly more per transaction. Marketing investment in these categories should focus on conversion quality over traffic volume.

5. Phase 3 — Customer Demand Analysis

5.1 Top 10 Categories by Unique Customers

This analysis counts the number of distinct customers who placed at least one order in each category, identifying which product types attract the widest customer base.

Table 6. Top 10 Product Categories by Unique Customers

Rank	Product Category	Unique Customers
1	Bed Bath Table	5,390
2	Health Beauty	5,024
3	Sports Leisure	4,383
4	Computers Accessories	3,837
5	Furniture Decor	3,684
6	Housewares	3,423
7	Watches Gifts	3,267
8	Telephony	2,416
9	Auto	2,240
10	Toys	~1,980

Figure 8. Top 10 Product Categories by Unique Customers

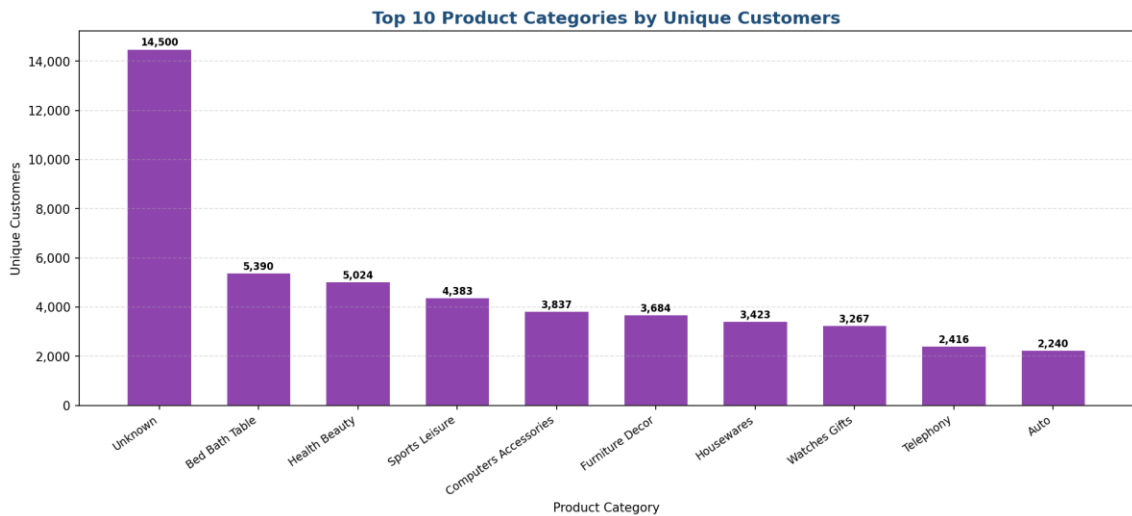


Figure 8: Bed Bath Table (5,390 customers) and Health Beauty (5,024) attract the widest individual customer base. The ranking mirrors the top-selling order list, confirming consistent demand across both metrics.

5.2 Top 10 Categories by Repeat Purchases

This analysis counts total purchase instances per category (including multiple purchases by the same customer), identifying categories that drive the highest repeat engagement.

Table 7. Top 10 Product Categories by Total Purchases (Including Repeats)

Rank	Product Category	Total Purchases
1	Bed Bath Table	6,725
2	Health Beauty	5,749
3	Sports Leisure	5,156
4	Furniture Decor	5,029
5	Computers Accessories	4,653
6	Housewares	4,282
7	Watches Gifts	3,649
8	Telephony	2,750
9	Garden Tools	2,712
10	Auto	2,629

Figure 9. Top 10 Product Categories by Repeat Purchases

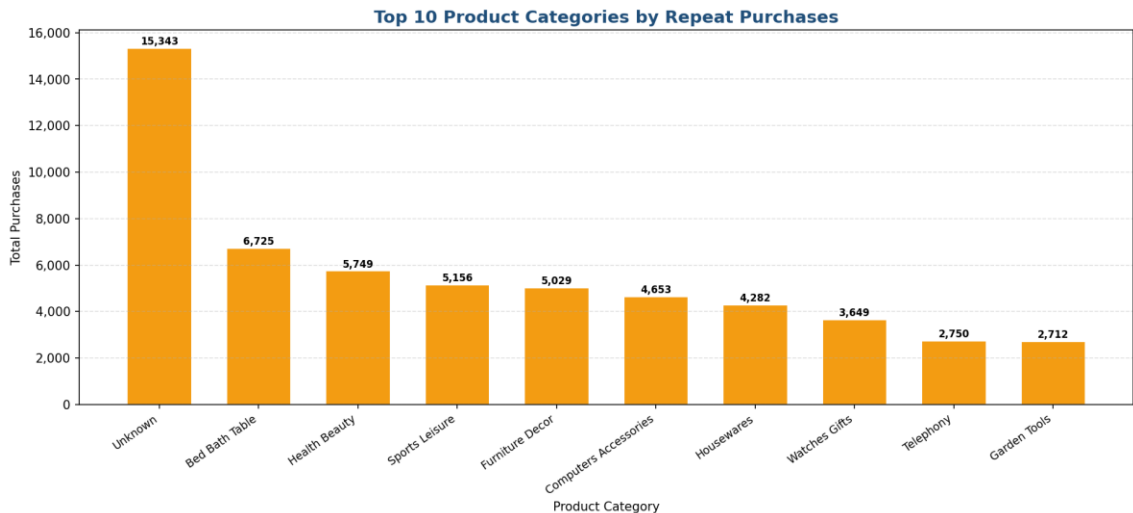


Figure 9: Bed Bath Table leads repeat purchases with 6,725 total instances — 23% more than its unique order count, indicating meaningful repeat purchasing behaviour in this category.

Comparing repeat purchase counts to unique order counts reveals the purchase frequency multiplier per category. Bed Bath Table (6,725 total purchases vs 5,473 unique orders) and Furniture Decor (5,029 total purchases vs 3,731 unique orders) show the strongest repeat multipliers, suggesting these categories attract customers who return for additional purchases within the same product area.

6. Phase 4 — Product Profitability Analysis

Figure 10. Product Profitability Scatter — Revenue vs Orders



Figure 10: Scatter plot of revenue vs orders per category. Top-right = high volume AND high revenue (ideal). Top-left = high revenue, low orders (premium). Bottom-right = high orders, low revenue (commodity).

The profitability scatter plot segments categories into four strategic quadrants based on their order volume and revenue performance:

- High Volume + High Revenue (top-right): Health Beauty, Bed Bath Table, Sports Leisure, Computers Accessories — these are the core platform performers driving both traffic and revenue.
- Premium Low-Volume (top-left): Watches Gifts, Cool Stuff, Auto — high revenue per transaction despite moderate order counts; these are premium categories requiring quality-focused customer experience.
- High Volume Low Revenue (bottom-right): Telephony, Toys — popular categories with moderate revenue per order; margin optimisation is important.
- Low Volume Low Revenue (bottom-left): Security And Services, Flowers, La Cuisine, Cds Dvds — candidates for catalogue review or targeted niche marketing.

7. Phase 5 — Geographic Demand Analysis

Product demand was mapped across Brazilian states by creating a cross-tabulation of customer state against the top 10 product categories. This identifies regional concentration and geographic demand patterns for the platform's best-selling categories.

Figure 11. Product Demand by State — Top 10 Categories Heatmap

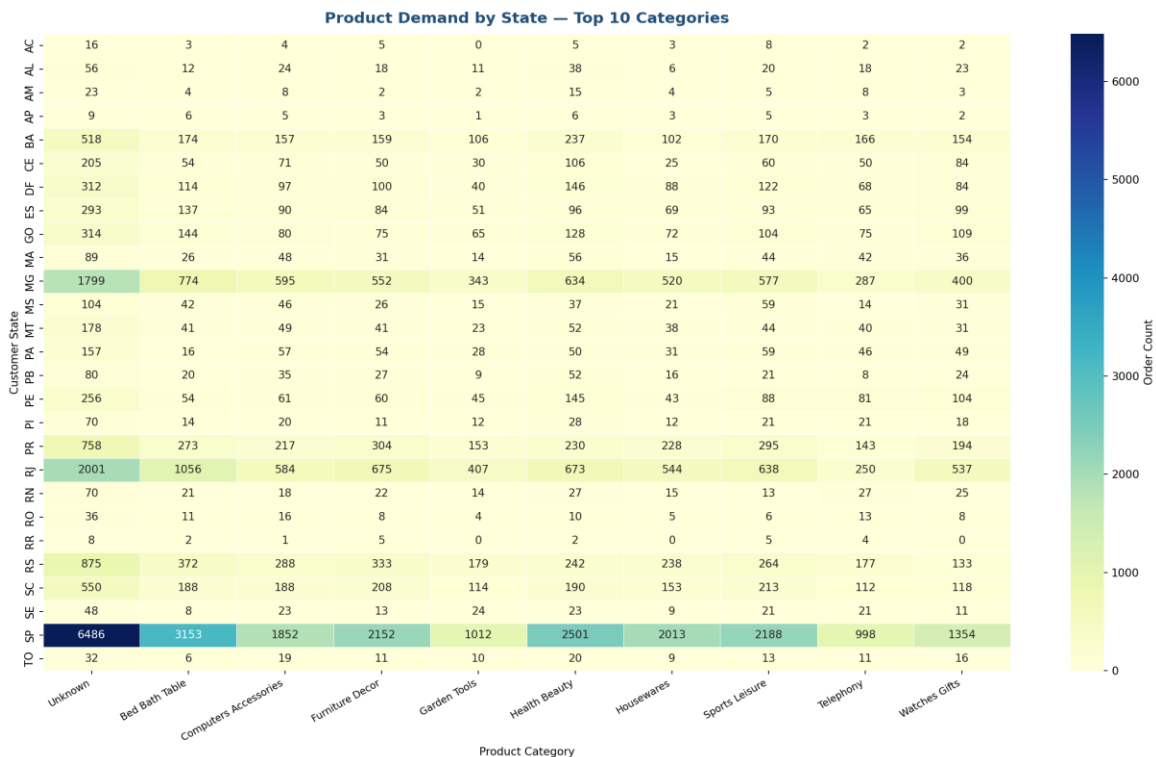


Figure 11: Heatmap of order counts per state per product category. SP dominates across all categories. Darker cells = higher demand. The concentration in SP, RJ, MG reflects the platform's geographic revenue distribution.

São Paulo (SP) dominates product demand across all top 10 categories, consistent with SP's 37%+ share of total platform revenue identified in earlier analyses. Key geographic observations from the product-state matrix:

- SP shows 500+ orders in Bed Bath Table, Health Beauty, and Sports Leisure — confirming these as core SP market categories.
- RJ and MG show strong demand in Bed Bath Table, Health Beauty, and Computers Accessories, mirroring the national category rankings.
- Northern and remote states (AC, AP, AM, RR) show very low demand across all categories, consistent with their low overall revenue contribution.
- BA, GO, and DF show moderate but consistent demand across most top 10 categories, representing active mid-tier markets.

8. Phase 6 — RFM Segment Product Preferences

The RFM Customer Segmentation file was merged with the product dataset on customer_unique_id to identify which product categories are preferred by each customer behavioural segment. This is one of the most actionable analyses in the product report, enabling targeted category marketing per segment.

8.1 Champions' Favorite Product Categories

Table 8. Champions' Top 10 Favorite Product Categories

Rank	Product Category	Purchases
1	Health Beauty	65
2	Watches Gifts	64
3	Housewares	53
4	Furniture Decor	44
5	Sports Leisure	41
6	Bed Bath Table	39
7	Computers Accessories	38
8	Cool Stuff	30
9	Auto	19
10	Toys	18

Figure 12. Champions' Top 10 Favorite Product Categories

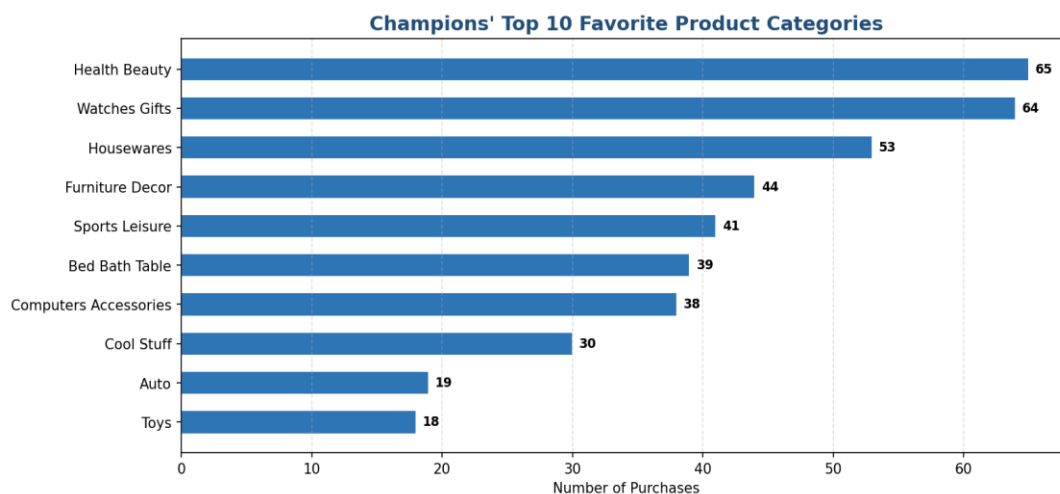


Figure 12: Champions favour Health Beauty (65) and Watches Gifts (64) — both premium and personal-care categories. This is consistent with Champions being high-spend, high-recency customers.

Champions — the platform's highest-value customers (429 customers averaging R\$257.80 each) — show clear preference for premium personal care and lifestyle categories. Health Beauty and Watches Gifts are nearly tied as their top two categories, confirming that this segment values quality and personal

presentation products. Marketing campaigns targeting Champions should emphasise premium product lines in these categories with exclusive or early-access offers.

8.2 Loyal Customers' Favorite Product Categories

Table 9. Loyal Customers' Top 10 Favorite Product Categories

Rank	Product Category	Purchases
1	Health Beauty	89
2	Bed Bath Table	81
3	Computers Accessories	77
4	Sports Leisure	75
5	Furniture Decor	63
6	Telephony	55
7	Housewares	50
8	Watches Gifts	43
9	Electronics	38
10	Auto	~30

Figure 13. Loyal Customers' Top 10 Favorite Product Categories

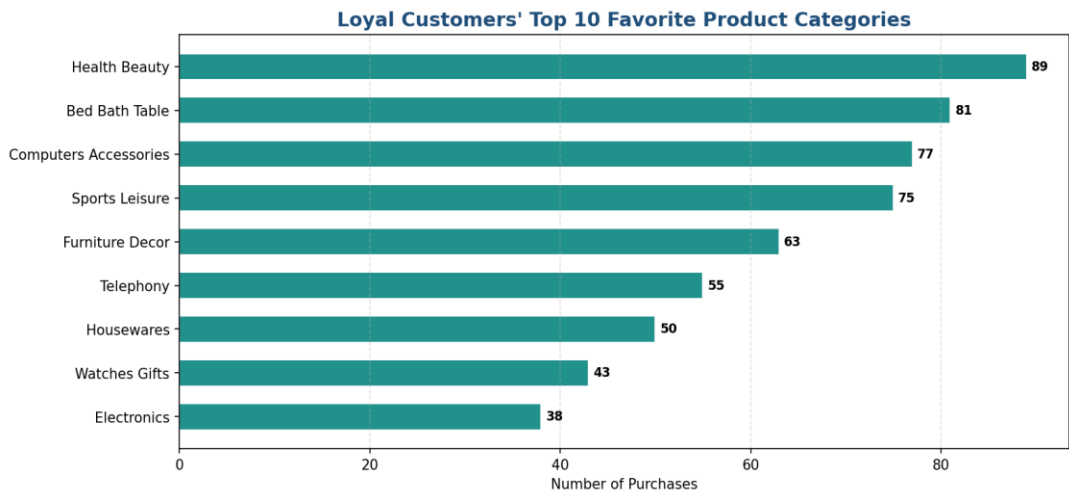


Figure 13: Loyal Customers show the highest absolute purchases in Health Beauty (89), reflecting their larger segment size (1,098 customers vs 429 Champions). Their preferences skew toward home and tech categories.

Loyal Customers (1,098 customers, avg R\$62.13) show a broader product preference profile than Champions, favouring both personal care (Health Beauty) and home/lifestyle categories (Bed Bath Table, Computers Accessories, Sports Leisure). The presence of Telephony and Electronics in their top 10 — absent from Champions' list — suggests that Loyal Customers are more tech-oriented buyers who purchase moderate-value everyday products. Cross-sell opportunities exist for upselling this segment into the premium categories that Champions prefer.

8.3 Product vs Customer Segment Heatmap

Figure 14. Customer Segment vs Product Category Heatmap (Top 15 Categories)

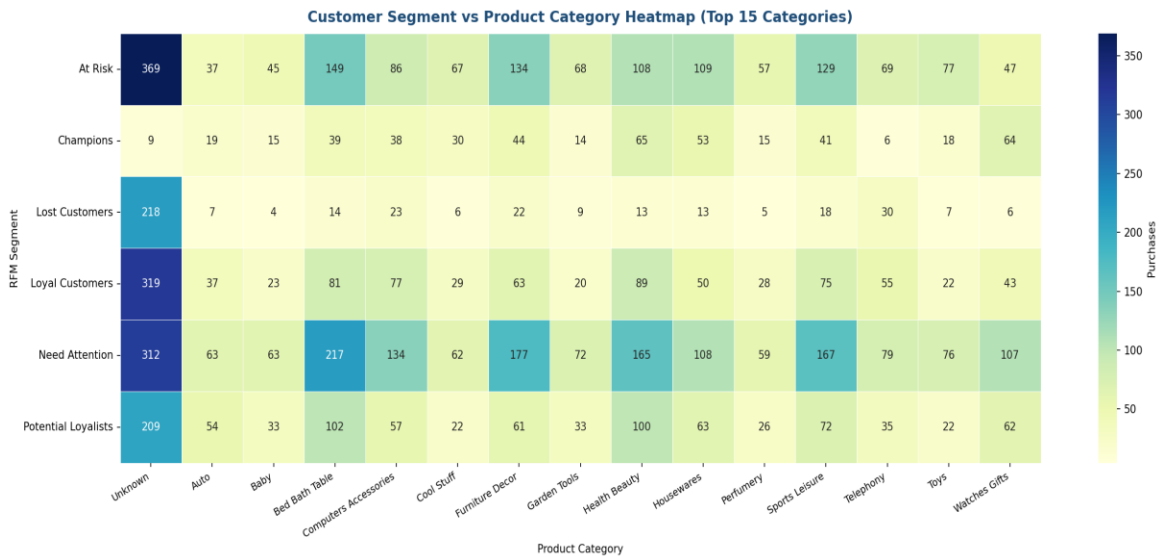


Figure 14: Heatmap of purchases by RFM segment and product category. Darker cells = higher purchase concentration. Need Attention (largest segment) dominates most category columns by volume.

Figure 15. Top 5 Product Categories per RFM Segment

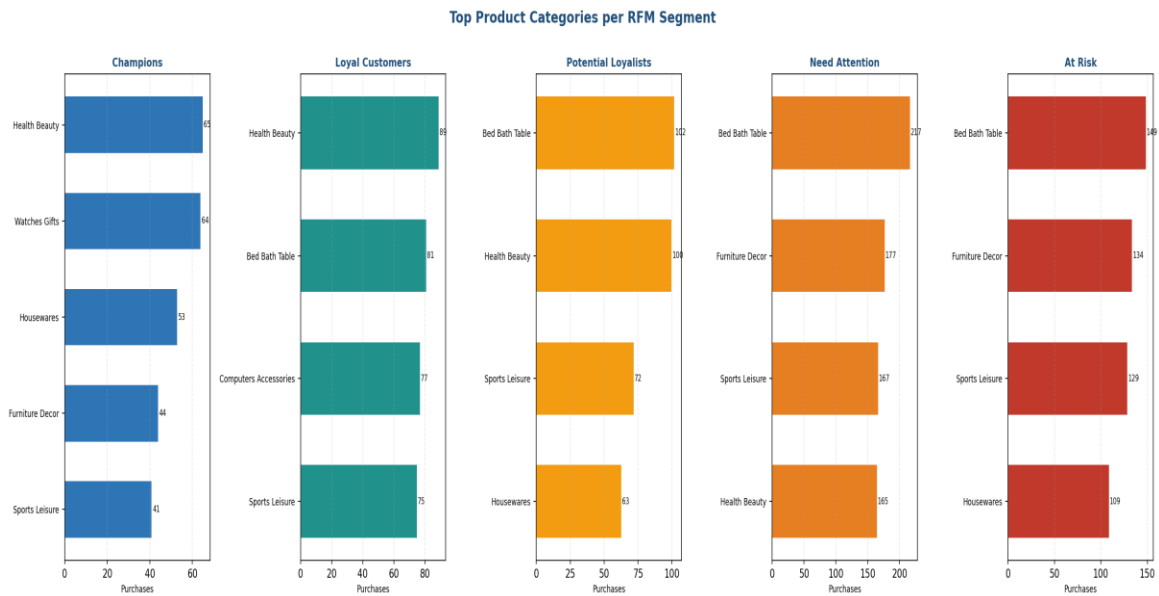


Figure 15: Side-by-side view of the top 5 product categories for each RFM segment. Champions and Loyal Customers share Health Beauty as #1; At Risk customers concentrate in Bed Bath Table; Need Attention spans diverse categories.

The segment-product heatmap reveals important behavioural distinctions across the customer lifecycle. Need Attention customers (1,870 customers) show broad category engagement with moderate volumes across all top categories — they were active buyers who have become less frequent. At Risk customers (1,492 customers) concentrate most heavily in Bed Bath Table, suggesting this may be a one-time purchase category driving subsequent inactivity. Champions show concentrated preference in premium personal care and lifestyle categories, enabling precise category-level VIP marketing.

9. Key Findings Summary

Table 10. Prioritised Key Findings

#	Finding Area	Finding	Business Impact
1	Revenue vs Volume Divergence	Watches Gifts ranks #1 by revenue (R\$743K) but only #7 by orders — highest AOV category on the platform	Very High
2	Dual-Leader Categories	Health Beauty and Bed Bath Table rank in the top 3 across ALL five dimensions: orders, revenue, customers, repeat purchases, and segment preference	Very High
3	Champions Premium Preference	Champions prefer Health Beauty (65) and Watches Gifts (64) — both high AOV categories; confirms premium positioning drives top-segment satisfaction	High
4	SP Geographic Concentration	SP dominates demand across all top 10 categories; 500+ orders per category for the top 3 categories in SP alone	High
5	Repeat Purchase Leaders	Bed Bath Table has a 23% repeat purchase premium over its unique order count; strongest repeat category on the platform	High
6	Low-Performing Tail	10 categories generate under R\$1,600 total revenue; Security And Services (R\$283) and Fashion Childrens Clothes (R\$420) are near-zero contributors	Moderate
7	Loyal vs Champion Divergence	Loyal Customers prefer Computers Accessories and Telephony (tech categories); Champions prefer Watches Gifts and Housewares — actionable cross-sell pathway	Moderate

10. Business Recommendations

Table 11. Prioritised Business Recommendations

Area	Recommendation	Priority
Protect Core Categories	Maintain deep stock, fast delivery, and high seller quality for Health Beauty and Bed Bath Table — the platform's dual leaders across volume, revenue, and customer reach. Any service degradation in these categories risks significant revenue loss.	Very High
Premium Category Investment	Invest in Watches Gifts category growth through curated seller selection, improved product photography, and targeted campaigns. It leads revenue despite ranking 7th by orders — AOV improvement campaigns here have high leverage.	Very High
Champions Personalisation	Launch Health Beauty and Watches Gifts exclusive product recommendations and early-access campaigns specifically for Champions. These are their confirmed top categories — personalisation converts at much higher rates.	High
Loyal Customer Cross-Sell	Use Loyal Customers' preference for Computers Accessories and Telephony as an entry point for cross-selling into Watches Gifts and Cool Stuff — the categories that drive Champions' higher average spend.	High
SP Market Deepening	SP shows 500+ orders per top category. Prioritise fast-delivery seller concentration in SP for Health Beauty, Bed Bath Table, and Sports Leisure — same-day or next-day delivery here would have the highest revenue impact.	High
Low-Tail Catalogue Review	Review the 10 categories generating under R\$1,600 total revenue. Evaluate whether to remove, merge, invest in, or leave these as-is. Security And Services (R\$283 total) should be considered for deactivation unless strategically required.	Moderate
Geographic Expansion	CE, PE, and BA show moderate but consistent demand across top categories. Targeted regional campaigns with improved delivery SLAs in these states could unlock the next tier of growth outside the SP/RJ/MG core.	Moderate

11. Limitations and Next Steps

The following limitations apply to this product analysis:

- 'Unknown' category (14,600 orders, highest in dataset) represents unclassified products. If these could be re-classified, the category rankings would shift materially.
- Revenue = price + freight_value; this does not include seller commissions or platform fees, so true profitability per category cannot be determined.
- Repeat purchase analysis counts all line-item purchases, not distinct re-purchase events. The true repeat-buyer rate within each category would require session-level or order-event-level analysis.
- RFM merge is left-join on customer_unique_id; customers without RFM matches show no segment label and are excluded from segment-level counts. This may slightly underrepresent segment preferences.
- Dataset covers Sep 2016 to Aug 2018 only. Category trends may have shifted significantly after this period.

Recommended next steps include: market basket analysis to identify which categories are frequently co-purchased, time-series category revenue forecasting, seller-level category quality scoring, and A/B testing of category page optimisations for the top revenue categories.

Appendix A. Analysis Summary

Table 12. Product Analysis Outputs Generated

Output	Description
Top 10 Selling Categories	Ranked by unique order count; Unknown excluded from table
Bottom 10 Selling Categories	Lowest order volume categories; identifies catalogue weaknesses
Top 10 Revenue Categories	Ranked by total revenue (price + freight_value)
Bottom 10 Revenue Categories	Lowest revenue categories with exact R\$ values
Revenue Contribution Pie	Share of top-10-category revenue visualised
Top 10 by Unique Customers	Broadest customer reach by category
Top 10 by Repeat Purchases	Total purchase instances per category including repeat orders
Average Order Value by Category	Mean revenue per order per category; identifies premium segments
Product Demand by State Heatmap	Cross-tab of state vs top 10 categories; geographic demand map
Champions' Product Preferences	Top 10 categories purchased by Champions segment
Loyal Customers' Preferences	Top 10 categories purchased by Loyal Customers segment
Segment vs Product Heatmap	Full crosstab of all 6 RFM segments vs top 15 categories
Profitability Scatter	Revenue vs Orders scatter to identify strategic quadrants
Top Products per Segment Panel	Side-by-side top-5 products for 5 RFM segments

*All analyses performed using Python (pandas, numpy, matplotlib, seaborn) in a Jupyter Notebook environment.
Datasets: analysis_data.csv (82,365 rows, Olist master dataset) and RFM_Customer_Segmentation.xlsx (6,263 customers).*